

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

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Unit I | Session 1 | SLO 2: Syntax and Semantics (Code: 1012)

Q: 1. Which of the following is not an element of a programming language?

a) Syntax b) Algorithm c) Semantics d) Data types

Ans: Answer: b) Algorithm

Explanation: An algorithm is a language-independent step-by-step logic or procedure to solve a problem. In contrast, Syntax, Semantics, and Data Types are core formal elements inherent to programming language specifications.

Q: 2. Identify the correct pair of syntax and semantics:

a) Brackets → Loop b) Colon → Meaning c) if-else → Logic rule d) Semicolon → Statement terminator

Ans: Answer: d) Semicolon → Statement terminator

Explanation: The semicolon (;) is the concrete syntax in C/C++/Java, and its formal semantic meaning is marking the termination of an execution statement.

Q: 3. True or False: The semantics of a language define its grammatical rules.

Ans: Answer: False

Explanation: Grammatical rules define how valid expressions and statements are composed, which is the domain of **Syntax**. **Semantics** defines the meaning, runtime behavior, and evaluation of syntactically valid constructs.

Q: 4. What is the element responsible for variable declaration and type enforcement?

Ans: Answer: Data Type System / Type System

Explanation: The type system governs variable declarations, sets memory allocation requirements, defines permissible operations, and enforces compile-time and runtime type safety constraints.

Q: 5. Which element determines how functions and control flows are structured?

Ans: Answer: Control Structures and Formal Grammar (Control Flow Constructs)

Explanation: Control structures (such as conditional branching, loops, and subroutine call/return mechanisms) dictate the execution path and instruction sequencing of a program.